Total No. of Printed Pages: 03

SUBJECT CODE NO: - RNP-8008
FACULTY OF SCIENCE AND TECHNOLOGY
F.E. (Group A) Mech / Civil (NEP) Sem - I
Examination June/July 2025
Engineering Mechanics

[Time: 3:00 Hours] **[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B
1) Use of non-programmable calculator is allowed
2) Q.no.1 is compulsory
3) Solve any four questions from Q.nos 2, 3, 4, 5, 6 and 7

SECTION A

Q1 Answer the following questions. **20**

1. The result of two or more forces acting on a body is called:
   a) Moment b) Resultant force
   c) Torque d) Equilibrium

2. Which of the following is a non-coplanar force system?
   a) All forces lie in the same plane
   b) Forces are at different angles in space
   c) All forces act parallel to one another
   d) All forces have a common point of action

3. What is the key characteristic of a couple in a force system?
   a) Two equal forces acting in the same direction
   b) Two forces of different magnitudes acting in the same direction
   c) Two equal, opposite forces separated by a distance
   d) Two forces acting at right angles to each other

4. Which method is used to analyze forces in a truss?
   a) Direct integration method b) Method of Joint
   c) Moment distribution method d) Newton's method

5. In a plane truss, if a member is under tension, the force in the member is:
   a) Positive b) Negative
   c) Zero d) Undefined

6. What is the centroid of a plane surface?
   a) The point where the area is maximum
   b) The point where the first moment of the area is zero
   c) The point of intersection of the axes of symmetry
   d) The geometric center of the area

7. Which of the following shapes has its centroid at the geometric center?
   a) Triangle b) Parabola
   c) Rectangle d) Semicircle

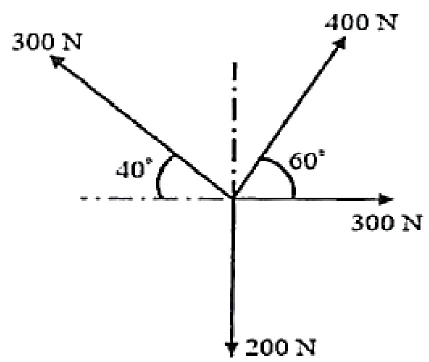
8. Which of the following is true about a system in equilibrium?
   a) The sum of all forces and the sum of all moments must be zero.
   b) Only the sum of all forces must be zero.
   c) Only the sum of all moments must be zero.
   d) None of the above.

9. The center of mass of an object is:
   a) The point where the total weight of the object acts.
   b) The point where all the mass of the object is concentrated.
   c) The point where all forces acting on the object balance each other.
   d) The point where the object has maximum density.

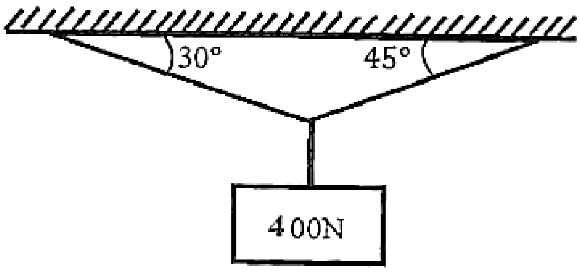
10. Which law of motion states that for every action, there is an equal and opposite reaction?
    a) Newton's First Law   b) Newton's Second Law
    c) Newton's Third Law   d) Law of Universal Gravitation

SECTION B

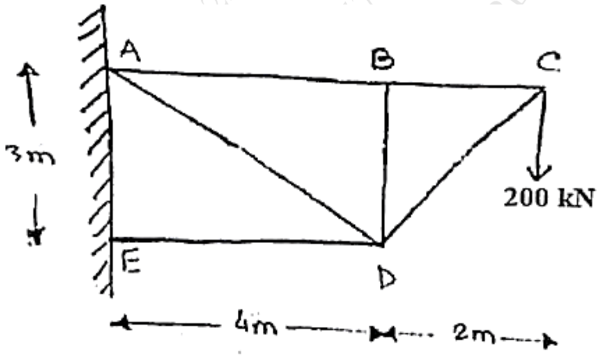
Q2 a) Find magnitude and direction of given concurrent force system. **05**



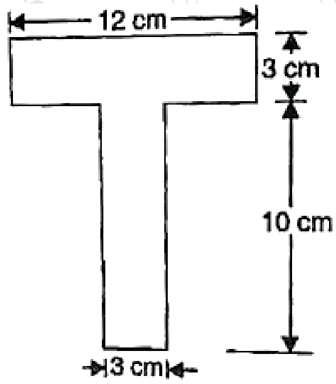
b) A weight of 400N is attached by two strings calculate the tension in the string **05**



Q3 Find out the magnitude of forces in given truss. **10**



Q4 Determine the centroid of T-section shown in figure below about X and Y axis. **10**



Q5 Displacement equation of a particle is given by, **10**

$$S = 25t + 6t^2 - 2t^3$$

Find:

a) Velocity and acceleration at start
b) Time when particle reaches its maximum velocity
c) The maximum velocity of the particle.

Q6 a) A ball is thrown with a velocity of 50 m/s at an angle of elevation of 40° with the horizontal. **05**

Find:

a) Maximum height reached by the ball
b) Time of flight
c) Range of ball

b) A bullet is fired from a gun with an initial velocity of 250 m/s to hit it a target. The target is located at a horizontal distance of 3500 m and 650 m above the gun. Determine minimum angle of projection so that the bullet will hit the target. **05**

Q7 An automobile of mass 1.2 tons moves down a 30° incline at a speed of 50 kmph, when the brakes are applied causing a constant braking force of 5 kN. Determine the distance travelled by the Automobile as it comes to rest. **10**